Software Requirements Specification (SRS)

AI-Powered Intelligent Interview Question Management System

Version: 1.0

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Table of Contents

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional, non-functional, and business requirements for the **AI-Powered Intelligent Interview Question Management System**.

The system is designed to provide organizations with a centralized platform for managing, organizing, and retrieving interview questions using Artificial Intelligence and Retrieval-Augmented Generation (RAG). It enables administrators to maintain a structured repository of interview questions categorized by organization, domain, technology, role, and difficulty level, while allowing end users to efficiently search and explore relevant questions through semantic search and AI-generated responses.

This document serves as the primary reference for stakeholders, business analysts, software architects, developers, testers, and project managers throughout the software development lifecycle. It establishes a common understanding of the system requirements and provides a foundation for system design, implementation, testing, deployment, and future maintenance.

1.2 Scope

The AI-Powered Intelligent Interview Question Management System is a web-based application that simplifies the management and retrieval of interview questions through Artificial Intelligence.

The system enables administrators to create, update, categorize, and manage interview questions using structured metadata such as organization, domain, technology, role, category, and difficulty level. End users can search the repository using natural language queries, apply metadata-based filters, and receive AI-generated responses along with contextually relevant interview questions.

The application leverages Retrieval-Augmented Generation (RAG), vector embeddings, semantic search, and Large Language Models (LLMs) to improve the accuracy and relevance of search results beyond traditional keyword-based search.

The proposed solution is intended to support interview preparation, internal recruitment processes, technical assessments, and enterprise knowledge management by providing intelligent and context-aware access to interview-related information.

1.3 Business Context

Organizations conducting technical recruitment often maintain interview questions across multiple spreadsheets, documents, and disconnected repositories. This fragmented approach makes question management difficult, introduces duplication, and limits the ability to quickly retrieve relevant interview questions during recruitment or candidate preparation.

The AI-Powered Intelligent Interview Question Management System addresses these challenges by providing a centralized and intelligent repository for interview questions. The platform combines structured metadata management with AI-powered semantic search, enabling users to retrieve relevant interview questions using natural language rather than relying solely on exact keyword matches.

By integrating Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs), the system delivers context-aware search results, AI-generated explanations, and related interview questions. This enhances recruiter productivity, improves candidate preparation, and ensures efficient knowledge management within the organization.

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2. Business Requirements

2.1 Business Objectives

The primary objective of the AI-Powered Intelligent Interview Question Management System is to establish a centralized platform for managing and retrieving interview questions across multiple organizations, technologies, and domains.

The business objectives are:

- Centralize interview questions into a single, searchable repository.
- Enable intelligent retrieval of interview questions using semantic search instead of traditional keyword matching.
- Improve candidate interview preparation by providing contextually relevant questions and AI-assisted responses.
- Reduce the effort required to organize and manage interview questions.
- Standardize question categorization across organizations, domains, technologies, roles, and difficulty levels.
- Ensure secure access to interview content through Role-Based Access Control (RBAC).
- Provide a scalable platform capable of supporting multiple organizations, interview domains, and future technology expansions.

2.2 Stakeholders

Stakeholder	Responsibility
Administrator	Manages organizations, domains, technologies, interview questions, users, and overall system configuration.
End User (Candidate/Employee)	Searches interview questions, applies filters, and utilizes AI-generated responses for interview preparation and learning.

Organization / Recruitment Team	Maintains interview question quality, categorization standards, and oversees the interview knowledge repository.

2.3 Business Rules

The system shall adhere to the following business rules:

- Every interview question shall be associated with an organization, domain, technology, role, and difficulty level.
- Only users with the Administrator role shall be authorized to create, modify, or delete interview questions and metadata.
- End users shall have read-only access to search and AI-assisted interview preparation features.
- Organization and domain filters shall be applied before presenting search results.
- AI-generated responses shall be generated only from context retrieved from the interview question repository.
- Duplicate interview questions for the same organization, domain, and technology shall not be permitted.
- All administrative operations shall be recorded for audit purposes.

2.4 Assumptions & Constraints

Assumptions

- The organization will maintain an up-to-date and validated interview question repository.
- Question metadata (organization, domain, technology, role, and difficulty) is accurately maintained to support semantic retrieval.
- External AI services required for response generation are available and accessible.

- User roles and permissions are provisioned through the organization's access management process.

Constraints

- The quality of AI-generated responses is dependent on the completeness and relevance of the stored interview question repository.
- Semantic search performance depends on the quality of generated vector embeddings and indexing strategy.
- The system relies on third-party AI services for response generation; service outages or API limitations may temporarily affect AI functionality.
- Changes to organization taxonomy, domains, or technologies require administrative updates to maintain metadata consistency.
- User authorization shall strictly follow the Role-Based Access Control (RBAC) policy implemented within the system.

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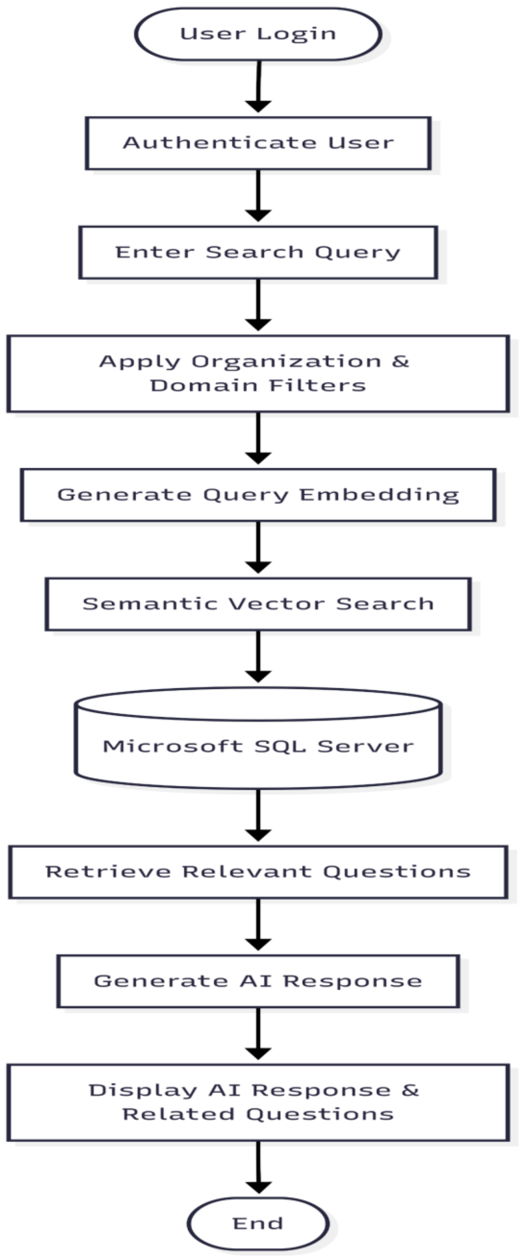
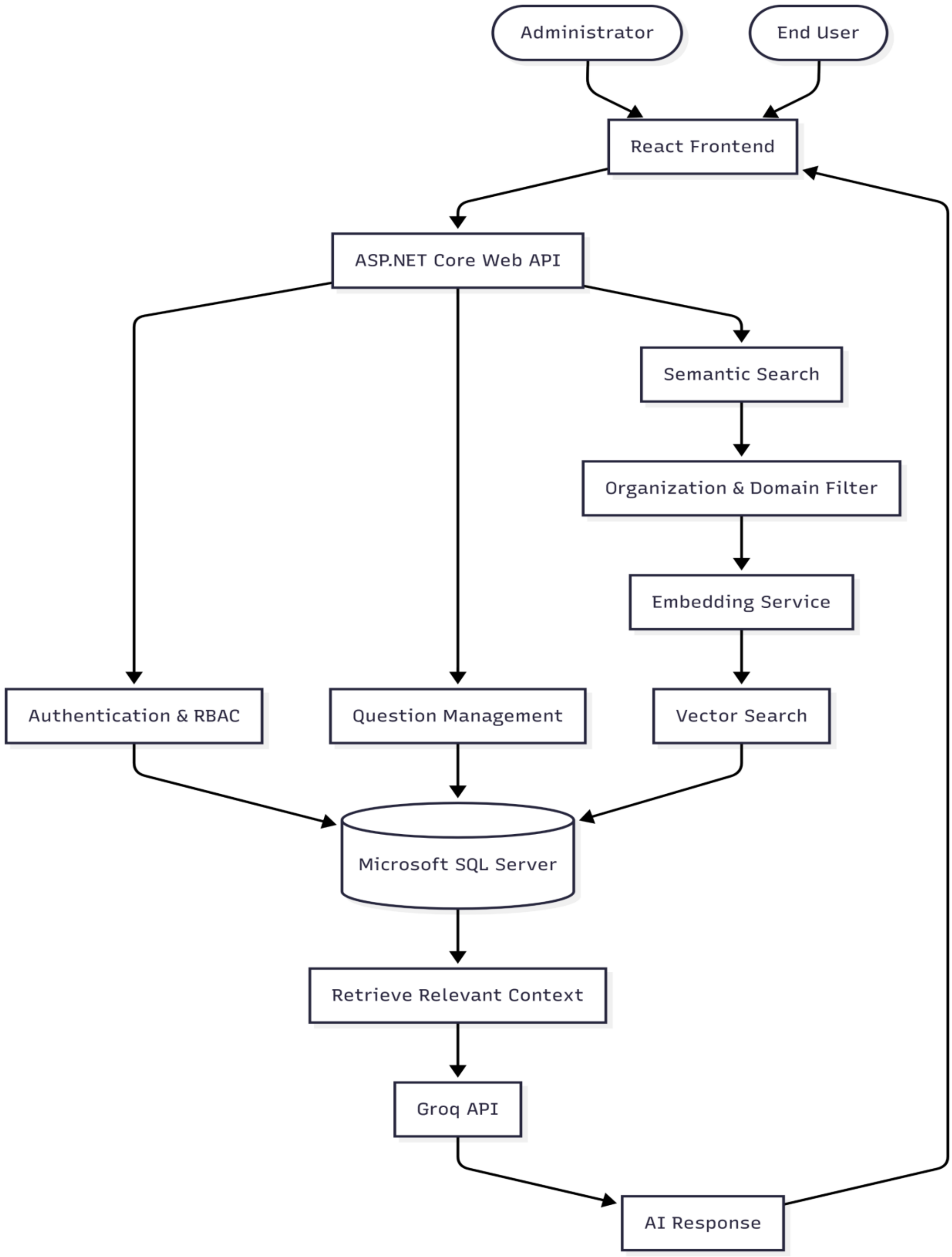
3. System Architecture

3.1 Data Flow



3.2 Architecture Diagram



4. User Roles & Role-Based Access Control (RBAC)

4.1 User Roles

The AI-Powered Intelligent Interview Question Management System implements Role-Based Access Control (RBAC) to ensure secure and authorized access to system resources. The platform defines two primary user roles:

Administrator

The Administrator is responsible for managing the interview question repository and maintaining the overall system. This role has full access to administrative functionalities, including question management, metadata management, and user administration.

End User

The End User is an individual preparing for interviews or accessing interview-related content. End users can search interview questions, apply organization and domain filters, and utilize AI-assisted search capabilities. They have read-only access to the interview repository and are not permitted to modify system data.

4.2 Permission List

Administrator

Authorized Operations

- Create Interview Questions
- Update Interview Questions
- Delete Interview Questions
- Manage Organizations
- Manage Domains
- Manage Technologies
- Manage Users
- Access Dashboard

- Perform Semantic Search
- View AI Responses

End User

Authorized Operations

- Search Interview Questions
- Apply Organization Filters
- Apply Domain Filters
- Perform Semantic Search
- View AI Responses
- View Related Questions

Restricted Operations

- Create Interview Questions
- Modify Interview Questions
- Delete Interview Questions
- Manage Organizations
- Manage Domains
- Manage Technologies
- Manage Users
- Access Administrative Dashboard

4.3 Authentication

The system shall authenticate users before granting access to application resources. User credentials are validated by the ASP.NET Core Web API, and authenticated users are issued a secure JSON Web Token (JWT). The token is required for accessing protected API endpoints throughout the application.

Authentication ensures that only verified users can access the system and establishes the identity of each user before authorization policies are applied.

4.4 Authorization

Authorization is implemented using Role-Based Access Control (RBAC). After successful authentication, the system evaluates the user's assigned role to determine the operations they are permitted to perform.

Administrative functions—including interview question management, organization management, domain management, technology management, and user management—are restricted exclusively to users assigned the Administrator role.

End Users are authorized to search, filter, and view interview questions and AI-generated responses but are restricted from performing any data modification or administrative operations.

The authorization mechanism ensures secure access control, protects system integrity, and enforces the principle of least privilege by granting users only the permissions necessary for their assigned responsibilities.

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5. Functional Requirements

FR-01 User Authentication

Description

The system shall provide a secure authentication mechanism that verifies the identity of users before granting access to protected resources. Only authenticated users shall be permitted to access application functionalities based on their assigned roles.

Actor

- Administrator
- End User

Preconditions

- The user must possess a valid account.
- The user must not already be authenticated.

Inputs

- Username or Email Address
- Password

Processing

1. The user submits valid login credentials.
2. The system validates the credentials.
3. Upon successful validation, the system authenticates the user.
4. The system determines the assigned user role.
5. A secure authentication session is established.

Outputs

- Successful login.
- Access granted according to the assigned user role.
- Authentication failure message for invalid credentials.

Postconditions

- The authenticated user can access authorized system resources.

Business Rules

- Only registered users can authenticate.
- Invalid credentials shall not grant system access.
- Authentication shall be completed before authorization is evaluated.

Acceptance Criteria

- Valid users can successfully log in.
- Invalid credentials return an authentication error.
- User role is identified after successful authentication.
- Unauthorized users cannot access protected resources.

FR-02 Question Management

Description

The system shall allow administrators to create, update, view, and delete interview questions while maintaining associated metadata such as organization, domain, technology, role, category, and difficulty level.

Actor

- Administrator

Preconditions

- Administrator must be authenticated.

Inputs

- Interview Question
- Answer
- Organization

- Domain
- Technology
- Role
- Category
- Difficulty Level

Processing

1. Administrator submits question details.
2. System validates mandatory fields.
3. System stores or updates the interview question.
4. Metadata is associated with the interview question.
5. Changes are committed to the database.

Outputs

- Question successfully created, updated, or deleted.
- Confirmation message.

Postconditions

- Repository reflects the latest interview question data.

Business Rules

- Only administrators may manage interview questions.
- Mandatory metadata must be provided.
- Duplicate questions should be prevented whenever applicable.

Acceptance Criteria

- Administrator can successfully perform Create, Read, Update, and Delete operations.
- Metadata is correctly stored with each question.
- Unauthorized users cannot perform management operations.

FR-03 Organization Management

Description

The system shall allow administrators to manage organizations used for categorizing interview questions. Administrators can create, update, view, and remove organizations to maintain a standardized repository.

Actor

- Administrator

Preconditions

- Administrator must be authenticated.

Inputs

- Organization Name
- Organization Description (Optional)

Processing

1. Validate organization details.
2. Check for duplicate organization names.
3. Store or update organization information.
4. Reflect changes in the repository.

Outputs

- Organization successfully created, updated, or deleted.

Acceptance Criteria

- Only administrators can manage organizations.
- Duplicate organization names are not permitted.

FR-04 Domain Management

Description

The system shall allow administrators to manage interview domains such as Frontend, Backend, Full Stack, AI/ML, DevOps, etc.

Actor

- Administrator

Preconditions

- Administrator must be authenticated.

Inputs

- Domain Name
- Domain Description (Optional)

Processing

1. Validate domain information.
2. Verify uniqueness.
3. Save or update the domain.

Outputs

- Domain successfully managed.

Acceptance Criteria

- Only administrators can manage domains.
- Duplicate domains are not allowed.

FR-05 Organization Filter

Description

The system shall allow users to filter interview questions based on the selected organization.

Actor

- Administrator

- End User

Inputs

- Organization Name

Outputs

- Interview questions belonging to the selected organization.

Acceptance Criteria

- Only questions belonging to the selected organization are displayed.

FR-06 Domain Filter

Description

The system shall allow users to filter interview questions based on the selected interview domain.

Actor

- Administrator
- End User

Inputs

- Domain Name

Outputs

- Interview questions belonging to the selected domain.

Acceptance Criteria

- Only questions belonging to the selected domain are displayed.

FR-07 Semantic Search

Description

The system shall support semantic search by understanding the contextual meaning of user queries instead of relying solely on keyword matching.

Actor

- Administrator
- End User

Inputs

- Natural Language Query

Processing

1. Convert the query into vector embeddings.
2. Perform similarity search.
3. Retrieve the most relevant interview questions.

Outputs

- Contextually relevant interview questions.

Acceptance Criteria

- Relevant questions are retrieved even when exact keywords are absent.
- Search results are ranked by semantic similarity.

FR-08 Keyword Classification

Description

The system shall analyse user queries and identify the associated organization, domain, or technology to improve search accuracy.

Actor

- Administrator
- End User

Inputs

- Search Query

Outputs

- Identified organization, domain, or technology.
- Enhanced search results.

Acceptance Criteria

- The system correctly identifies relevant metadata from user queries.

FR-09 AI Response Generation

Description

The system shall generate intelligent responses using Retrieval-Augmented Generation (RAG) by combining retrieved interview questions with a Large Language Model.

Actor

- Administrator
- End User

Inputs

- User Query
- Retrieved Context

Processing

1. Retrieve relevant interview questions.
2. Provide contextual information to the LLM.
3. Generate an AI-assisted response.

Outputs

- AI-generated response.
- Supporting interview questions.

Acceptance Criteria

- AI responses are generated using retrieved repository context.
- Responses are relevant to the user's query.

FR-10 Recommendation Engine

Description

The system shall recommend related interview questions based on semantic similarity to help users explore additional relevant topics.

Actor

- Administrator
- End User

Inputs

- Current Search Result

Outputs

- List of related interview questions.

Acceptance Criteria

- Recommendations are contextually related to the selected question.

FR-11 Admin Dashboard

Description

The system shall provide administrators with a centralized dashboard for managing interview questions, organizations, domains, users, and repository information.

Actor

- Administrator

Preconditions

- Administrator must be authenticated.

Outputs

- Dashboard displaying management options and repository overview.

Acceptance Criteria

- Only administrators can access the dashboard.
- All administrative modules are accessible from a single interface.

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6. Use Cases

6.1 End User Use Cases

Use Case ID

UC-01

Use Case Name

Search and Retrieve Interview Questions

Primary Actor

End User

Description

This use case describes how an end user searches for interview questions using natural language queries and metadata-based filters. The system retrieves contextually relevant interview questions and generates AI-assisted responses.

Preconditions

- The user is authenticated.
- The interview question repository contains searchable data.

Trigger

The user submits a search query through the application.

Main Flow

1. The user logs into the system.
2. The user enters a search query.
3. The user optionally selects an organization and/or domain filter.
4. The system analyzes the search query.
5. The system performs semantic search.
6. Relevant interview questions are retrieved.
7. The retrieved context is provided to the AI model.

8. The system displays AI-generated responses along with related interview questions.

Alternative Flow

- If no relevant interview questions are found, the system displays an appropriate notification.
- If the AI service is temporarily unavailable, the system displays the retrieved interview questions without an AI-generated response.

Postconditions

- Relevant interview questions are presented to the user.
- AI-generated responses are displayed when available.

6.2 Administrator Use Cases

Use Case ID

UC-02

Use Case Name

Manage Interview Question Repository

Primary Actor

Administrator

Description

This use case describes how an administrator manages interview questions, organizations, and domains through the administrative interface.

Preconditions

- The administrator is authenticated.
- The administrator has sufficient privileges.

Trigger

The administrator selects a management operation.

Main Flow

1. The administrator logs into the system.
2. The administrator accesses the dashboard.
3. The administrator selects the required management module.
4. The administrator creates, updates, or deletes interview questions.
5. The administrator manages organizations and domains.
6. The system validates the submitted information.
7. The repository is updated successfully.

Alternative Flow

- If validation fails, the system displays the corresponding error message.
- If duplicate information is detected, the system prevents the operation.

Postconditions

- Repository data is updated.
- Changes are immediately available for future searches.

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7. Non-Functional Requirements

7.1 Performance

The system shall provide fast and efficient retrieval of interview questions while maintaining a responsive user experience.

Requirements:

- The system shall process user requests with minimal latency.
- Search results shall be displayed within an acceptable response time under normal operating conditions.
- The system shall support concurrent user requests without significant degradation in performance.
- Database queries shall be optimized to improve retrieval efficiency.

7.2 Scalability

The system shall be designed to accommodate future growth in data volume and user activity.

Requirements:

- The system shall support the addition of new organizations, domains, technologies, and interview questions without requiring architectural changes.
- The application shall be capable of handling increasing numbers of concurrent users.
- The database shall support efficient storage and retrieval as the interview repository expands.

7.3 Availability

The system shall remain accessible to authorized users whenever required.

Requirements:

- The application shall be available during scheduled operational hours.
- Planned maintenance activities shall be communicated in advance.

- Temporary service interruptions shall be minimized.

7.4 Reliability

The system shall consistently perform its intended functions without data loss or unexpected failures.

Requirements:

- The application shall maintain data integrity during Create, Update, and Delete operations.
- System failures shall not corrupt stored interview questions or metadata.
- Appropriate error handling mechanisms shall be implemented to prevent application crashes.

7.5 Security

The system shall protect application data and restrict access to authorized users.

Requirements:

- User authentication shall be required before accessing protected resources.
- Role-Based Access Control (RBAC) shall regulate access to system functionality.
- Administrative operations shall be restricted to authorized users.
- Sensitive user information shall be securely stored and protected.

7.6 Maintainability

The application shall be designed to facilitate future enhancements and maintenance activities.

Requirements:

- The system shall follow a modular software architecture.

- Source code shall be organized according to established development standards.
- New features shall be incorporated with minimal impact on existing functionality.
- Software documentation shall be maintained throughout the project lifecycle.

7.7 Usability

The application shall provide an intuitive and user-friendly interface for both administrators and end users.

Requirements:

- Navigation shall be simple and consistent across all modules.
- Search functionality shall be easily accessible.
- Error messages shall clearly describe the encountered issue.
- The user interface shall provide meaningful feedback for user actions.

7.8 Logging & Monitoring

The system shall maintain operational logs to support monitoring, troubleshooting, and auditing activities.

Requirements:

- Authentication events shall be recorded.
- Administrative operations shall be logged.
- Application errors and exceptions shall be captured for analysis.
- Logs shall support troubleshooting and system maintenance.